

Alpamayo1.5 Finetune Migration - Change Report

1. Overview

This report documents all changes made to migrate the finetune training pipeline (SFT and RL) from the alpamayo (alpamayo_r1) repository to the alpamayo1.5 (alpamayo1_5) repository. The goal is to enable alpamayo1.5 models to be fine-tuned using the same SFT/RL training infrastructure that was originally built for alpamayo_r1.

Key differences between the two repositories:

- Package name: alpamayo_r1 -> alpamayo1_5
- Alpamayo1.5 adds include_camera_ids, include_frame_nums config fields
- Alpamayo1.5 base_model has generate_text() method (new)
- Alpamayo1.5 uses _padding_N tokens instead of image_pre_tkn, meta_action_start etc.
- Alpamayo1.5 has nav_utils.py for navigation features
- Alpamayo1.5 uses Qwen3-VL instead of Qwen2.5-VL as VLM backbone

2. New Source Modules (src/alpamayo1_5/)

These modules were copied from alpamayo_r1 because the finetune training code depends on them, but they were missing in the alpamayo1.5 repository. All internal references were updated from alpamayo_r1 to alpamayo1_5.

[src/alpamayo1_5/common/__init__.py](#)

Reason: Common utilities package init

[src/alpamayo1_5/common/config_utils.py](#)

Reason: Hydra config saving/loading utilities used by SFT training

[src/alpamayo1_5/common/constants.py](#)

Reason: Camera name constants and mappings used by data preprocessing

[src/alpamayo1_5/common/distributed.py](#)

Reason: Distributed training utilities (NCCL init, rank helpers)

[src/alpamayo1_5/common/logging.py](#)

Reason: RankedLogger for distributed-aware logging in training

[src/alpamayo1_5/common/misc.py](#)

Reason: Miscellaneous utilities (seed_everything, param counting, pformat)

[src/alpamayo1_5/common/wandb_utils.py](#)

Reason: Weights & Biases initialization for experiment tracking

[src/alpamayo1_5/chat_template/__init__.py](#)

Reason: Chat template package init

[src/alpamayo1_5/chat_template/conversation.py](#)

Reason: VLM conversation template builder for data preprocessing

[src/alpamayo1_5/processor/__init__.py](#)

Reason: Processor package init

[src/alpamayo1_5/processor/qwen_processor.py](#)

Reason: Qwen VLM processor with tokenization, collation, and chat template

[src/alpamayo1_5/utils/__init__.py](#)

Reason: Utils package init

[src/alpamayo1_5/utils/get_label_mask.py](#)

Reason: Label mask generation for supervised fine-tuning loss

[src/alpamayo1_5/data/pai.py](#)

Reason: PAI dataset class for loading Physical AI AV data

[src/alpamayo1_5/data/pai_utils.py](#)

Reason: PAI dataset utilities and local data interface

[src/alpamayo1_5/metrics/distance_metrics.py](#)

Reason: Trajectory distance metrics (ADE, FDE, collision rate)

[src/alpamayo1_5/metrics/metric_api.py](#)

Reason: Metric API classes (ReasoningSampler, DistanceMetrics)

[src/alpamayo1_5/metrics/metric_runner.py](#)

Alpamayo1.5 Finetune Migration - Change Report

Reason: MetricRunner for evaluation metric computation

[src/alpamayo1_5/metrics/metric_utils.py](#)

Reason: Metric helper functions (summarize, prefix)

3. New Finetune SFT Files (finetune/sft/)

The entire SFT (Supervised Fine-Tuning) directory was copied from alpamayo and adapted. All imports were changed from alpamayo_r1 to alpamayo1_5. Class names were updated (TrainableAlpamayoR1 -> TrainableAlpamayo1_5, AlpamayoR1 -> Alpamayo1_5).

[finetune/sft/train_hf.py](#)

Reason: Main SFT training entry point. Imports updated to alpamayo1_5.common

[finetune/sft/trainer.py](#)

Reason: Custom HuggingFace Trainer with LR multiplier support

[finetune/sft/evaluate_hf.py](#)

Reason: SFT evaluation script. Import updated: sft_alpamayo_r1 -> sft_alpamayo1_5

[finetune/sft/models/sft_base_model.py](#)

Reason: TrainableReasoningVLA base model for SFT. Imports updated to alpamayo1_5

[finetune/sft/models/sft_alpamayo1_5.py](#)

Reason: TrainableAlpamayo1_5 model (renamed from sft_alpamayo_r1.py). Class AlpamayoR1 -> Alpamayo1_5, config AlpamayoR1Config -> Alpamayo1_5Config

[finetune/sft/configs/sft_base.yaml](#)

Reason: Base SFT config. _target_ paths updated to alpamayo1_5.processor/data/metrics

[finetune/sft/configs/sft_stage1.yaml](#)

Reason: Stage 1 SFT config (VLM fine-tuning)

[finetune/sft/configs/sft_stage2.yaml](#)

Reason: Stage 2 SFT config (expert model fine-tuning)

[finetune/sft/configs/vla_processor.yaml](#)

Reason: VLA processor config with alpamayo1_5 target

[finetune/sft/configs/models/ar1_base.yaml](#)

Reason: Base model config pointing to TrainableReasoningVLA

[finetune/sft/configs/models/ar1_expert.yaml](#)

Reason: Expert model config. _target_ updated to sft_alpamayo1_5.TrainableAlpamayo1_5

[finetune/sft/configs/deepspeed/zero2.json](#)

Reason: DeepSpeed ZeRO-2 configuration for distributed training

[finetune/sft/configs/wandb/default.yaml](#)

Reason: Default W&B logging config

[finetune/sft/configs/__init__.py](#)

Reason: Configs package init

4. New Finetune RL Files (finetune/rl/)

The entire RL (Reinforcement Learning with GRPO) directory was copied from alpamayo and adapted. All imports changed from alpamayo_r1 to alpamayo1_5. The RL pipeline uses Cosmos-RL framework with vLLM for rollout generation.

[finetune/rl/__init__.py](#)

Reason: RL package init

[finetune/rl/state.py](#)

Reason: Global state management for Cosmos-RL. Imports updated to alpamayo1_5

[finetune/rl/launcher.py](#)

Reason: Cosmos-RL launch utilities

[finetune/rl/base_cosmos_wrapper.py](#)

Reason: Base Cosmos model wrapper

[finetune/rl/base_data_packer.py](#)

Reason: Base data packer for RL training

Alpamayo1.5 Finetune Migration - Change Report

[finetune/rl/base_dataset.py](#)

Reason: Base dataset class for RL

[finetune/rl/base_trainer.py](#)

Reason: Base GRPO trainer for Alpamayo

[finetune/rl/models/__init__.py](#)

Reason: Models package init

[finetune/rl/models/_spec.py](#)

Reason: ModelSpec definition for Cosmos-RL components

[finetune/rl/models/reasoning_vla/__init__.py](#)

Reason: ReasoningVLA model package init

[finetune/rl/models/reasoning_vla/base_model.py](#)

Reason: RLWrapperReasoningVLA. Imports updated to alpamayo1_5

[finetune/rl/models/reasoning_vla/config.py](#)

Reason: RLWrapperReasoningVLAConfig with include_camera_ids/frame_nums

[finetune/rl/models/reasoning_vla/cosmos_wrapper.py](#)

Reason: Cosmos BaseModel wrapper with FSDP2 sharding

[finetune/rl/models/reasoning_vla/data_packer.py](#)

Reason: Data packer bridging samples to vLLM prompts. Uses alpamayo1_5 SPECIAL_TOKENS

[finetune/rl/models/reasoning_vla/rollout.py](#)

Reason: vLLM rollout engine. Uses alpamayo1_5 SPECIAL_TOKENS

[finetune/rl/models/reasoning_vla/trainer.py](#)

Reason: ReasoningVLA GRPO trainer registered with Cosmos

[finetune/rl/models/reasoning_vla/vllm_wrapper.py](#)

Reason: vLLM model wrapper for ReasoningVLA inference

[finetune/rl/models/reasoning_vla/weight_mapper.py](#)

Reason: Weight name mapping between policy/rollout/HF formats

[finetune/rl/models/reasoning_vla/alpamayo_cosmos_rl_post_training_entry.py](#)

Reason: RL entry point. hydra_config_name=alpamayo1_5_rvla_rl_pai

[finetune/rl/models/reasoning_vla/alpamayo_cosmos_rl_post_training_entry_local_test.py](#)

Reason: Local test RL entry point

[finetune/rl/rewards/aggregated_reward.py](#)

Reason: Aggregated reward function (traj L2 + comfort)

[finetune/rl/rewards/comfort_reward.py](#)

Reason: Comfort reward (jerk, lateral acceleration)

[finetune/rl/rewards/traj_reward.py](#)

Reason: Trajectory L2 distance reward

[finetune/rl/utils/checkpoint.py](#)

Reason: Checkpoint utilities

[finetune/rl/utils/fsdp.py](#)

Reason: FSDP2 sharding utilities for visual tower and LM layers

[finetune/rl/utils/logit_processor.py](#)

Reason: Custom logit processors for generation

[finetune/rl/utils/trajectory_decode.py](#)

Reason: Decode rollout text to trajectory. Uses alpamayo1_5 tokens

[finetune/rl/utils/weight_loading.py](#)

Reason: DTensor weight loading utilities for FSDP2

[finetune/rl/prefetch/__init__.py](#)

Reason: Prefetch package init

[finetune/rl/prefetch/dataset.py](#)

Reason: Node-level prefetch dataset wrapper

[finetune/rl/prefetch/server.py](#)

Reason: Node-local prefetch server for RL data loading

Alpamayo1.5 Finetune Migration - Change Report

[finetune/rl/prefetch/shm.py](#)

Reason: Shared memory utilities for prefetch

[finetune/rl/prefetch/test_prefetch.py](#)

Reason: Prefetch E2E tests. Path comment updated to alpamayo1_5_release

[finetune/rl/hydra_configs/alpamayo1_5_rvla_rl_pai.yaml](#)

Reason: Hydra config for RL on PAI. All `_target_` use alpamayo1_5

[finetune/rl/hydra_configs/alpamayo1_rvla_rl_pai.yaml](#)

Reason: Alternative Hydra config (no camera_ids/frame_nums)

[finetune/rl/toml/alpamayo_rvla_rl_local_test.toml](#)

Reason: TOML config for local RL testing

[finetune/rl/run_rl_local_test_signal_node.sh](#)

Reason: Shell script updated: paths changed from alpamayo_r1_release to alpamayo1_5_release

[finetune/rl/notebooks/rl_checkpoint_inference.ipynb](#)

Reason: Inference notebook. Imports updated alpamayo_r1 -> alpamayo1_5

[finetune/rl/README.md](#)

Reason: RL documentation

[finetune/rl/assets/alpamayo_rl_framework.png](#)

Reason: RL framework diagram

[finetune/rl/assets/local_training_reward_curves.png](#)

Reason: Training reward curves reference

5. Modified Existing Files

The following existing file was modified to support the finetune training pipeline.

pyproject.toml

Changes:

1. Build system: `uv_build` -> `hatchling` (to support multi-package wheel builds)
Reason: Need to ship both `src/alpamayo1_5` and `finetune/rl` as installable packages
2. Added `[tool.hatch.build.targets.wheel]` with `packages=[src/alpamayo1_5, finetune/rl]`
Reason: Enable 'import rl' after pip install, matching alpamayo's setup
3. Added `[tool.hatch.build.targets.sdist]` with `include_paths`
Reason: Ensure source distribution includes training code
4. Added training dependencies: `cosmos-rl`, `colorlog`, `rich`, `wandb`, `torchmetrics`, `deepspeed`, `vllm`
Reason: Required by SFT training (`deepspeed`, `wandb`) and RL training (`cosmos-rl`, `vllm`)
5. Added `[tool.uv.sources]` `cosmos-rl` git source
Reason: `cosmos-rl` is not on PyPI, must be installed from GitHub

6. Compatibility Verification

The following compatibility points were verified:

- [OK] `ReasoningVLA.forward()` signature is compatible between `alpamayo_r1` and `alpamayo1_5`
- [OK] `TrajectoryFusionMixin.fuse_traj_tokens()` interface is identical
- [OK] `sample_trajectories_from_data_with_vlm_rollout()` exists in `Alpamayo1_5` with matching signature
- [OK] `SPECIAL_TOKENS` total count is identical (29 tokens)
- [OK] Critical tokens (`traj_future_start/end`, `cot_start/end`, `image_pad`) exist in both
- [OK] Token IDs are position-aligned (`_padding_N` replaces old unused tokens)
- [OK] `RLWrapperReasoningVLAConfig` includes `include_camera_ids`, `include_frame_nums`

Alpamayo1.5 Finetune Migration - Change Report

[OK] All alpamayo_r1 references removed (verified: 0 matches in grep)

[OK] All alpamayo1_5 imports point to existing modules

[NOTE] `construct_meta_action()` in `conversation.py` references `SPECIAL_TOKENS['meta_action_start/end']` which are renamed to `_padding_2/_padding_3` in `alpamayo1_5`. This function is NOT called by any current training config, so it poses no runtime risk.

7. Summary Statistics

New source modules added: 19 files (`src/alpamayo1_5/`)

New SFT finetune files: 14 files (`finetune/sft/`)

New RL finetune files: 38 files (`finetune/rl/`)

Modified existing files: 1 file (`pyproject.toml`)

Total files changed: 72 files

All alpamayo_r1 references: 0 remaining (fully cleaned)